

Effect sizes — formula sheet

FRB-CESAB training course · Wednesday 9:00 · keep it at hand during the exercises

Convention: T = intervention (e.g. intercropping), C = comparator (e.g. monoculture); $\bar{X}$ mean, SD standard deviation, n sample size, v sampling variance. Weight of a study in a meta-analysis: $w = 1/v$ (fixed effect) or $w = 1/(v + \tau^2)$ (random effects).

Quantity	Formula
Mean difference	$MD = \bar{X}_T - \bar{X}_C, \quad v = \frac{SD_T^2}{n_T} + \frac{SD_C^2}{n_C}$
Pooled SD	$SD_p = \sqrt{\frac{(n_T - 1)SD_T^2 + (n_C - 1)SD_C^2}{n_T + n_C - 2}}$
Hedges' g	$g = J \frac{\bar{X}_T - \bar{X}_C}{SD_p}, \quad v_g = \frac{1}{n_T} + \frac{1}{n_C} + \frac{g^2}{2(n_T + n_C)}$
Correction J ($m = n_T + n_C - 2$)	exact: $J = \frac{\Gamma(m/2)}{\sqrt{m/2} \Gamma((m-1)/2)}$ (used by escalc); approx.: $J \approx 1 - \frac{3}{4m-1}$
InRR (Hedges et al. 1999)	$\ln RR = \ln(\bar{X}_T / \bar{X}_C), \quad v = \frac{SD_T^2}{n_T \bar{X}_T^2} + \frac{SD_C^2}{n_C \bar{X}_C^2}$
Bias-corrected InRR (Lajeunesse 2015)	$\ln RR^\Delta = \ln RR + \frac{1}{2} \left(\frac{SD_T^2}{n_T \bar{X}_T^2} - \frac{SD_C^2}{n_C \bar{X}_C^2} \right),$ $v^\Delta = v + \frac{1}{2} \left(\frac{SD_T^4}{n_T^2 \bar{X}_T^4} + \frac{SD_C^4}{n_C^2 \bar{X}_C^4} \right)$
Geary's rule	mean / SE ≥ 3 (i.e. mean $\times \sqrt{n} / SD \geq 3$) in both groups; small n (Lajeunesse 2015): $\frac{\bar{X}}{SD} \frac{4n^{3/2}}{1+4n} \geq 3$
Log odds ratio (2x2 table $a \ b \ / \ c \ d$)	$\ln \frac{ad}{bc}, \quad v = \frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d}$ (add 0.5 to each cell if one is 0)
Fisher's z	$z = \frac{1}{2} \ln \frac{1+r}{1-r}, \quad v_z = \frac{1}{n-3}$
SD from SE	$SD = SE \times \sqrt{n}$
SD from a 95% CI	$SD = \sqrt{n} \times (\text{upper} - \text{lower}) / (2 \times 1.96)$ (≈ 3.92 only for large n ; with $n = 4$ use 6.37)
SD from median, IQR, range	estimators of Wan et al. (2014)
log OR $\rightarrow d$ (assumes a logistic latent trait)	$d = \ln OR \sqrt{3/\pi}, \quad v_d = v_{\ln OR} 3/\pi^2$
$r \rightarrow d$ (point-biserial r , two groups)	$d = \frac{2r}{\sqrt{1-r^2}}, \quad v_d = \frac{4v_r}{(1-r^2)^3}$; back: $r = d/\sqrt{d^2+4}$ (equal groups)
Back to %	$100 (e^{\ln RR} - 1)$

Which metric? Positive variable with a true zero, proportional effect (abundance, biomass, yield) $\rightarrow$ InRR. Scores or indices with an arbitrary zero, different scales $\rightarrow$ Hedges' g. Yes/no outcome $\rightarrow$ log odds ratio. Correlation $\rightarrow$ Fisher's z.

In R: `metafor::escalc(measure = "ROM" | "SMD" | "MD" | "OR" | "ZCOR", ...)` returns `yi` (effect size) and `vi` (sampling variance). A zero mean gives `NA`, a zero SD gives `vi = 0`: check both.

Bias-corrected InRR in R (not built into `escalc`):

```
tT <- sd1^2 / (n1 * m1^2); tC <- sd2^2 / (n2 * m2^2)
yi_c <- log(m1 / m2) + 0.5 * (tT - tC)
vi_c <- tT + tC + 0.5 * (tT^2 + tC^2)
```

Missing SDs for InRR (Nakagawa et al. 2023): replace the unknown CV^2/n terms by the n -weighted mean CV^2 of the studies with SDs. Conversions: only for a minority of studies, flagged, with a sensitivity analysis without them (Borenstein et al. 2021). Never convert InRR $\leftrightarrow$ g.

References

- Borenstein, M., L. V. Hedges, J. P. T. Higgins, and H. R. Rothstein. 2021. Introduction to meta-analysis. Second edition. Wiley, Chichester.
- Hedges, L. V., J. Gurevitch, and P. S. Curtis. 1999. [The meta-analysis of response ratios in experimental ecology](#). Ecology 80:1150–1156.
- Lajeunesse, M. J. 2015. [Bias and correction for the log response ratio in ecological meta-analysis](#). Ecology 96:2056–2063.
- Nakagawa, S., D. W. A. Noble, M. Lagisz, R. Spake, W. Viechtbauer, and A. M. Senior. 2023. [A robust and readily implementable method for the meta-analysis of response ratios with and without missing standard deviations](#). Ecology Letters 26:232–244.
- Wan, X., W. Wang, J. Liu, and T. Tong. 2014. [Estimating the sample mean and standard deviation from the sample size, median, range and/or interquartile range](#). BMC Medical Research Methodology 14:135.